

Discrete Mathematics and Graph Theory

Module 2 (CO2) — Combinatorics

Questions with Detailed Step-by-Step Solutions

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Every question solved from zero to the final answer — what is used, why it is used, and every small step explained

Each solution below is written so that a beginner can follow it: first *what the question asks*, then *which idea or formula is used and why*, then the *complete step-by-step working*, and finally the *answer*. Part A questions carry 2 marks; Part B questions carry 7 marks.

Part A — Short Questions (2 Marks)

Q1. (2 marks) State the Pigeonhole principle. Give an example.

Solution — full explanation.

What the question asks: state a counting rule and illustrate it.

The principle. If $k + 1$ (or more) objects are placed into k boxes, then **at least one box must contain two or more objects**.

Why is this true (the intuition)? Suppose every box were allowed at most one object. Then k boxes could hold at most k objects. But we have $k + 1$ objects — one more than the boxes can hold with one each. So the extra object must share a box with another. That is exactly why some box ends up with two.

Generalized form. If N objects go into k boxes, then some box contains at least $\lceil N/k \rceil$ objects (the ceiling means “round up”, because a box cannot hold a fraction of an object).

Example (birthdays). There are 366 possible birthdays (boxes) and 367 people (objects). Since $367 > 366$, by the principle at least two people share the same birthday. (Here $k = 366$, so $k + 1 = 367$.)

One more example. Among 100 people, there are only 12 birth-months, so some month contains at least $\lceil 100/12 \rceil = 9$ people.

Final answer. The Pigeonhole principle: $k + 1$ objects in k boxes $\Rightarrow$ some box holds ≥ 2 ; in general N objects in k boxes $\Rightarrow$ some box holds $\geq \lceil N/k \rceil$. Example: of 367 people, two share a birthday.

Q2. (2 marks) Use the Pigeonhole Principle to prove: in any group of 13 people, at least two were born in the same month.

Solution — full explanation.

What the question asks: prove a guaranteed “matching” among 13 people.

Identify boxes and objects. The 12 months are the **boxes** ($k = 12$); the 13 people are the **objects** ($N = 13$).

Apply the principle. If all 12 months had at most one person, then at most 12 people could be placed — one per month. But we have 13 people, which is more than 12. So at least one month must contain two or more people.

In formula. The guaranteed number in some month is $\lceil N/k \rceil = \lceil 13/12 \rceil = 2$.

Final answer. Hence at least two of the 13 people were born in the same month. ■

Q3. (2 marks) If 6 colours are used to paint 37 homes, show that at least 7 homes are of the same colour.

Solution — full explanation.

What the question asks: show a colour must be used on many homes.

Identify boxes and objects. The 6 colours are the **boxes** ($k = 6$); the 37 homes are the **objects** ($N = 37$).

Apply the generalized principle. Some colour is used on at least $\lceil 37/6 \rceil$ homes.

Compute. $37/6 = 6.16\dots$, and rounding up gives 7. So some colour is used on at least 7 homes.

Why rounding up? Homes are counted in whole numbers. If every colour were used on at most 6 homes, the total would be at most $6 \times 6 = 36$ homes — but we have 37. So at least one colour must go up to 7.

Final answer. At least 7 homes are painted the same colour. ■

Q4. (2 marks) How many solutions does $x + y + z = 17$ have, where x, y, z are non-negative integers?

Solution — full explanation.

What the question asks: count the triples (x, y, z) of whole numbers $(0, 1, 2, \dots)$ adding to 17.

Which idea is used and why. This is a “distribute identical items” problem. We use the **stars and bars** method. The formula for the number of non-negative integer solutions of $x_1 + x_2 + \dots + x_n = r$ is

$$\binom{n + r - 1}{r}.$$

Why this formula? Think of 17 identical stars “★” and 2 bars “|” (2 bars split the stars into 3 groups for x, y, z). Example: ★★|★★★★|★... means $x = 2, y = 4$, and the rest to z . Every arrangement of 17 stars and 2 bars gives exactly one solution, and every solution gives one arrangement. So we just arrange $17 + 2 = 19$ symbols in a line, choosing where the 2 bars go.

Compute. Number of ways to place 2 bars among 19 positions:

$$\binom{19}{2} = \frac{19 \times 18}{2} = 171.$$

Final answer. There are **171** solutions.

Q5. (2 marks) How many solutions are there of $x + y + z = 17$ subject to $x \geq 1, y \geq 2, z \geq 3$?

Solution — full explanation.

What the question asks: same as Q4, but each variable has a minimum value.

The trick. Remove the forced minimums so the variables become non-negative again. Substitute

$$x' = x - 1, \quad y' = y - 2, \quad z' = z - 3,$$

so that $x', y', z' \geq 0$. Substituting into the equation:

$$(x' + 1) + (y' + 2) + (z' + 3) = 17 \Rightarrow x' + y' + z' = 17 - 6 = 11.$$

Apply stars and bars (as in Q4) to $x' + y' + z' = 11$ with $n = 3, r = 11$:

$$\binom{3 + 11 - 1}{11} = \binom{13}{11} = \binom{13}{2} = 78.$$

Why subtract 6? The total “already used up” by the minimums is $1 + 2 + 3 = 6$; the remaining $17 - 6 = 11$ is distributed freely.

Final answer. There are **78** solutions.

Q6. (2 marks) Find the generating function for the sequence $0, 1, 2, 3, 4, \dots$

Solution — full explanation.

What the question asks: write the sequence as the coefficients of a power series.

Set up. Here the r -th term is $a_r = r$, so the generating function is

$$G(x) = \sum_{r \geq 0} a_r x^r = \sum_{r \geq 0} r x^r = 0 + 1x + 2x^2 + 3x^3 + \dots$$

Which standard result is used and why. We know $\sum_{r \geq 0} x^r = \frac{1}{1-x}$. Differentiating both sides

with respect to x gives $\sum_{r \geq 1} r x^{r-1} = \frac{1}{(1-x)^2}$. Multiplying by x turns x^{r-1} into x^r :

$$\sum_{r \geq 0} r x^r = x \cdot \frac{1}{(1-x)^2} = \frac{x}{(1-x)^2}.$$

Final answer. The generating function is $G(x) = \frac{x}{(1-x)^2}$.

Q7. (2 marks) Find the generating function for the sequence $1, 1, 1, 1, \dots$

Solution — full explanation.

What the question asks: encode the constant sequence as a power series.

Set up. Here $a_r = 1$ for every r , so

$$G(x) = 1 + x + x^2 + x^3 + \dots$$

Which result is used and why. This is a geometric series with first term 1 and ratio x ; its sum (for $|x| < 1$) is

$$1 + x + x^2 + \dots = \frac{1}{1-x}.$$

Final answer. The generating function is $G(x) = \frac{1}{1-x}$.

Q8. (2 marks) How many permutations can be formed from the letters of the word LEVEL?

Solution — full explanation.

What the question asks: count the distinct arrangements of the letters L, E, V, E, L.

Observe the repeats. LEVEL has 5 letters, but not all are different: **L** appears twice, **E** appears twice, and V appears once.

Idea. If all 5 letters were different, the number of arrangements would be $5! = 120$. But swapping the two L's gives the *same* word (it is not a new arrangement), and swapping the two E's also gives the same word. So $5!$ over-counts.

Correct it. The two L's can be rearranged among themselves in $2!$ ways (which we must not count as different), and similarly the two E's in $2!$ ways. So divide:

$$\frac{5!}{2!2!} = \frac{120}{2 \times 2} = \frac{120}{4} = 30.$$

Final answer. 30 distinct permutations.

Q9. (2 marks) Calculate the number of 3-combinations from a set of 5 distinct elements.

Solution — full explanation.

What the question asks: count the ways to choose 3 items out of 5, where order does not matter.

Which formula and why. A “combination” means an unordered selection, so we use

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Compute. With $n = 5, r = 3$:

$$\binom{5}{3} = \frac{5!}{3!2!} = \frac{120}{6 \times 2} = \frac{120}{12} = 10.$$

Final answer. There are **10** 3-combinations.

Q10. (2 marks) How many ways can 8 people be arranged in a line if two specific people must stand together?

Solution — full explanation.

What the question asks: count line arrangements of 8 people with a “must be adjacent” condition.

The block idea. When two people must stay together, tie them into a single “block”. Now instead of 8 separate people we arrange **7 units**: the block plus the other 6 people.

Arrange the units. The 7 units can be ordered in $7! = 5040$ ways.

Arrange inside the block. The two people inside the block can swap places in $2! = 2$ ways (person A first or person B first).

Combine (product rule). For each of the $7!$ unit-orders there are 2 internal orders:

$$7! \times 2! = 5040 \times 2 = 10080.$$

Final answer. **10,080** arrangements.

Q11. (2 marks) How many ways can 5 people be seated in a circle?

Solution — full explanation.

What the question asks: count circular (round-table) seatings.

Why a circle differs from a line. In a circle, rotating everyone to the next seat gives the *same* seating (nobody’s neighbours change). A line has a definite left and right end, but a circle has none.

The fix. Fix one person in a seat to break the rotational symmetry; then arrange the remaining $n - 1$ people. This is why the formula for n people in a circle is $(n - 1)!$.

Compute. With $n = 5$:

$$(5 - 1)! = 4! = 4 \times 3 \times 2 \times 1 = 24.$$

Final answer. **24** ways.

Q12. (2 marks) What is the maximum number of groups into which you can distribute 30 students such that at least one group must have at least 5 students?

Solution — full explanation.

What the question asks: the largest number of groups k for which it is *guaranteed* that some

group has ≥ 5 students.

Which idea and why. Use the generalized pigeonhole principle: putting 30 students into k groups guarantees a group with at least $\lceil 30/k \rceil$ students.

Set the condition. We need $\lceil 30/k \rceil \geq 5$. The ceiling is ≥ 5 exactly when

$$\frac{30}{k} > 4 \Rightarrow k < \frac{30}{4} = 7.5.$$

Pick the largest integer. Since k must be a whole number less than 7.5, the largest is $k = 7$.

Check. $k = 7$: $\lceil 30/7 \rceil = \lceil 4.28 \rceil = 5 \geq 5 \checkmark$. $k = 8$: $\lceil 30/8 \rceil = \lceil 3.75 \rceil = 4 < 5 \times$.

Final answer. The maximum is **7** groups.

Q13. (2 marks) **Find the number of derangements of the set $\{1, 2, 3, 4\}$.**

Solution — full explanation.

What the question asks: count rearrangements of 1,2,3,4 in which *no* number stays in its own place.

What a derangement is. A derangement is a permutation with no element in its original position. For example, 2143 is a derangement (1 is not in position 1, 2 not in position 2, etc.), but 1234 is not.

Which formula and why. By inclusion–exclusion (subtract arrangements that fix at least one element), the number of derangements of n objects is

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!} \right).$$

Compute for $n = 4$.

$$D_4 = 4! \left(1 - \frac{1}{1} + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} \right) = 24 \left(\frac{24 - 24 + 12 - 4 + 1}{24} \right) = 24 \cdot \frac{9}{24} = 9.$$

Final answer. There are **9** derangements.

Q14. (2 marks) **Solve the recurrence relation $a_n = 2a_{n-1}$ with $a_0 = 1$.**

Solution — full explanation.

What the question asks: find a direct formula for a_n .

Unroll the recurrence. Each term is twice the previous one:

$$a_1 = 2a_0 = 2, \quad a_2 = 2a_1 = 4, \quad a_3 = 2a_2 = 8, \quad a_4 = 16, \dots$$

See the pattern. The values are 1, 2, 4, 8, 16, $\dots$, i.e. powers of 2. In general, starting from $a_0 = 1$ and doubling n times gives 2^n .

Final answer. $a_n = 2^n$.

Q15. (2 marks) **Solve the recurrence relation $a_n = 3a_{n-1} + 2$ with $a_0 = 2$.**

Solution — full explanation.

What the question asks: find a direct formula for a_n .

The plan. For a linear recurrence with a constant term, write the solution as

$$a_n = a_n^h + a_n^p$$

(a homogeneous part plus a particular part).

Particular part a_n^p . Since the forcing term “2” is a constant, assume $a_n^p = C$ (a constant). Substituting into $a_n = 3a_{n-1} + 2$:

$$C = 3C + 2 \Rightarrow -2C = 2 \Rightarrow C = -1.$$

So $a_n^p = -1$.

Homogeneous part a_n^h . Solve $a_n = 3a_{n-1}$, i.e. $\alpha = 3$, giving $a_n^h = A \cdot 3^n$.

Combine. $a_n = A \cdot 3^n - 1$.

Use the initial condition $a_0 = 2$. $A \cdot 3^0 - 1 = A - 1 = 2 \Rightarrow A = 3$.

Answer. $a_n = 3 \cdot 3^n - 1 = 3^{n+1} - 1$.

Final answer. $a_n = 3^{n+1} - 1$. (Check: $a_0 = 3 - 1 = 2$ ✓, and $a_1 = 3a_0 + 2 = 8 = 9 - 1$ ✓.)

Q16. (2 marks) Find the particular solution of the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 7^n$.

Solution — full explanation.

What the question asks: find only the particular solution a_n^p (the part that responds to the right-hand term 7^n).

Step 1 — find the characteristic roots. For the homogeneous part, solve the characteristic equation

$$\alpha^2 - 5\alpha + 6 = 0 \Rightarrow (\alpha - 2)(\alpha - 3) = 0 \Rightarrow \alpha = 2, 3.$$

Step 2 — choose the form of a_n^p . The right-hand side is 7^n , so $\beta = 7$. Because 7 is **not** a characteristic root (the roots are 2 and 3), we take the simple trial form

$$a_n^p = A \cdot 7^n.$$

Step 3 — substitute and solve for A . Put $a_n^p = A7^n$, $a_{n-1}^p = A7^{n-1}$, $a_{n-2}^p = A7^{n-2}$:

$$A7^n - 5A7^{n-1} + 6A7^{n-2} = 7^n.$$

Divide by 7^n (or multiply through by 7^2): $A - \frac{5}{7}A + \frac{6}{49}A = 1$. Multiplying by 49: $49A - 35A + 6A = 49 \Rightarrow 20A = 49 \Rightarrow A = \frac{49}{20}$.

Final answer. $a_n^p = \frac{49}{20}7^n$. (The full solution is $a_n = A_12^n + A_23^n + \frac{49}{20}7^n$.)

Q17. (2 marks) Find the particular solution of the recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 4^n$.

Solution — full explanation.

What the question asks: find only the particular solution a_n^p .

Step 1 — characteristic roots.

$$\alpha^2 - 7\alpha + 10 = 0 \Rightarrow (\alpha - 2)(\alpha - 5) = 0 \Rightarrow \alpha = 2, 5.$$

Step 2 — choose the form. The right-hand side is 4^n , so $\beta = 4$. Since 4 is **not** a root (roots are 2, 5), take $a_n^p = A \cdot 4^n$.

Step 3 — substitute and solve.

$$A4^n - 7A4^{n-1} + 10A4^{n-2} = 4^n \Rightarrow A \left(1 - \frac{7}{4} + \frac{10}{16}\right) = 1.$$

Compute the bracket: $1 - \frac{7}{4} + \frac{10}{16} = \frac{16-28+10}{16} = \frac{-2}{16} = -\frac{1}{8}$. So

$$-\frac{1}{8}A = 1 \Rightarrow A = -8.$$

Final answer. $a_n^p = -8 \cdot 4^n$.

Part B — Long Questions (7 Marks)

Q18. (7 marks) **A committee of 7 has to be formed from 9 boys and 4 girls. In how many ways can this be done when the committee consists of (i) exactly 3 girls, (ii) at least 3 girls, (iii) at most 3 girls?**

Solution — full explanation.

What the question asks: count committees of size 7 (order does not matter — it is a selection), under three different conditions on the number of girls.

Setup and idea. Total people = $9 + 4 = 13$. Choosing a committee means picking some girls and the rest boys; the choices are independent, so we multiply (product rule). We use combinations $\binom{n}{r}$ because order within the committee does not matter.

(i) Exactly 3 girls. If there are exactly 3 girls, the other $7 - 3 = 4$ members are boys. Choose 3 girls from 4 *and* 4 boys from 9:

$$\binom{4}{3} \cdot \binom{9}{4} = 4 \times 126 = 504.$$

(ii) At least 3 girls. “At least 3” means 3 girls *or* 4 girls (a 7-member committee cannot have more than 4 girls since only 4 exist).

- Exactly 3 girls: $\binom{4}{3} \binom{9}{4} = 504$.
- Exactly 4 girls: then $7 - 4 = 3$ boys, so $\binom{4}{4} \binom{9}{3} = 1 \times 84 = 84$.

Add (they are mutually exclusive): $504 + 84 = 588$.

(iii) At most 3 girls. “At most 3” means 0, 1, 2 or 3 girls. It is easier to count the complement: total committees minus those with 4 girls.

$$\text{Total} = \binom{13}{7} = 1716, \quad \text{with 4 girls} = \binom{4}{4} \binom{9}{3} = 84.$$

$$\text{Answer} = 1716 - 84 = 1632.$$

Final answer. (i) **504**, (ii) **588**, (iii) **1632**.

Q19. (7 marks) **In how many ways can 7 people be seated such that two specific people are never adjacent?**

Solution — full explanation.

What the question asks: count seatings of 7 in a row where a particular pair is *not* side by side.

Strategy (complement). It is easier to count all seatings, then subtract the “bad” ones where the pair *is* adjacent.

Total seatings. 7 people in a row: $7! = 5040$.

Bad seatings (pair adjacent). Treat the two people as one block: now arrange 6 units (the block + 5 others) $\Rightarrow 6! = 720$; the block’s two people can swap in $2!$ ways. So

$$6! \times 2! = 720 \times 2 = 1440.$$

Subtract. Never adjacent = total – adjacent:

$$5040 - 1440 = 3600.$$

Final answer. 3600 ways.

Q20. (7 marks) **How many integers between 1 and 500 (inclusive) are (a) divisible by at least one of 3, 5, 7? (b) divisible by 3 and 5 but not by 7?**

Solution — full explanation.

What the question asks: count numbers in 1..500 satisfying divisibility conditions.

Tool. For divisibility counts we use the floor function: the number of integers ≤ 500 divisible by d is $\lfloor 500/d \rfloor$.

Collect the basic counts.

$$\lfloor 500/3 \rfloor = 166, \quad \lfloor 500/5 \rfloor = 100, \quad \lfloor 500/7 \rfloor = 71.$$

Pairwise (divisible by both). Divisible by both 3 and 5 means divisible by $\text{lcm} = 15$; 3 and 7 by 21; 5 and 7 by 35:

$$\lfloor 500/15 \rfloor = 33, \quad \lfloor 500/21 \rfloor = 23, \quad \lfloor 500/35 \rfloor = 14.$$

All three. Divisible by 3, 5 and 7 means divisible by 105:

$$\lfloor 500/105 \rfloor = 4.$$

(a) By inclusion–exclusion. Add the singles, subtract the pairs, add back the triple:

$$|A \cup B \cup C| = 166 + 100 + 71 - 33 - 23 - 14 + 4 = 271.$$

(b) Divisible by 3 and 5 but not 7. Divisible by 3 and 5 means divisible by 15: that gives 33 numbers. From these, remove those also divisible by 7, i.e. divisible by 105: that gives 4 numbers. Hence

$$33 - 4 = 29.$$

Final answer. (a) 271, (b) 29.

Q21. (7 marks) **How many integer solutions exist for $x_1 + x_2 + x_3 = 10$ where $x_1 \geq 2$ and $x_2, x_3 \geq 0$?**

Solution — full explanation.

What the question asks: count non-negative solutions with one variable having a minimum.

Remove the minimum. Let $x'_1 = x_1 - 2 \geq 0$. Substituting $x_1 = x'_1 + 2$:

$$(x'_1 + 2) + x_2 + x_3 = 10 \Rightarrow x'_1 + x_2 + x_3 = 10 - 2 = 8.$$

Apply stars and bars. For $x'_1 + x_2 + x_3 = 8$ with $n = 3$ variables:

$$\binom{3 + 8 - 1}{8} = \binom{10}{8} = \binom{10}{2} = 45.$$

Final answer. 45 solutions.

Q22. (7 marks) **Use the generating function to solve $a_n = 3a_{n-1} + 2$, $a_0 = 1$.**

Solution — full explanation.

What the question asks: solve the recurrence using generating functions (a systematic method).

Step 1 — define $G(x)$ and set up the sum. Let $G(x) = \sum_{n \geq 0} a_n x^n$. Multiply the recurrence by x^n and sum for $n \geq 1$:

$$\sum_{n \geq 1} a_n x^n = 3 \sum_{n \geq 1} a_{n-1} x^n + 2 \sum_{n \geq 1} x^n.$$

Step 2 — rewrite each sum in terms of $G(x)$. We use

$$\sum_{n \geq 1} a_n x^n = G(x) - a_0 = G(x) - 1, \quad \sum_{n \geq 1} a_{n-1} x^n = xG(x), \quad \sum_{n \geq 1} x^n = \frac{x}{1-x}.$$

So

$$G - 1 = 3xG + \frac{2x}{1-x}.$$

Step 3 — solve for $G(x)$.

$$G(1-3x) = 1 + \frac{2x}{1-x} = \frac{1-x+2x}{1-x} = \frac{1+x}{1-x} \Rightarrow G(x) = \frac{1+x}{(1-x)(1-3x)}.$$

Step 4 — partial fractions. Write $\frac{1+x}{(1-x)(1-3x)} = \frac{A}{1-x} + \frac{B}{1-3x}$. Then $A(1-3x) + B(1-x) = 1+x$.

- Put $x = 1$: $A(1-3) = 2 \Rightarrow -2A = 2 \Rightarrow A = -1$.
- Put $x = \frac{1}{3}$: $B(1-\frac{1}{3}) = 1 + \frac{1}{3} \Rightarrow \frac{2}{3}B = \frac{4}{3} \Rightarrow B = 2$.

So $G(x) = \frac{-1}{1-x} + \frac{2}{1-3x}$.

Step 5 — read off the coefficients. Using $\frac{1}{1-x} = \sum x^n$ and $\frac{1}{1-3x} = \sum 3^n x^n$:

$$a_n = -1 \cdot 1 + 2 \cdot 3^n = 2 \cdot 3^n - 1.$$

Final answer. $a_n = 2 \cdot 3^n - 1$.

Q23. (7 marks) Find the solution of the recurrence relation $a_n - 3a_{n-1} = 2(3^n)$.

Solution — full explanation.

What the question asks: solve a non-homogeneous recurrence whose right side is a power of 3.

Step 1 — homogeneous part. Solve $a_n - 3a_{n-1} = 0$: characteristic $\alpha - 3 = 0 \Rightarrow \alpha = 3$, so $a_n^h = A_1 \cdot 3^n$.

Step 2 — choose the form of a_n^p . The right side is $2 \cdot 3^n$, so $\beta = 3$. But 3 is a characteristic root. When β is a root, the simple trial $A \cdot 3^n$ would clash with the homogeneous solution, so we multiply by n :

$$a_n^p = An \cdot 3^n.$$

Step 3 — substitute and solve for A . With $a_n^p = An3^n$ and $a_{n-1}^p = A(n-1)3^{n-1}$:

$$An3^n - 3A(n-1)3^{n-1} = 2 \cdot 3^n.$$

Since $3 \cdot 3^{n-1} = 3^n$, this becomes $An3^n - A(n-1)3^n = 2 \cdot 3^n$. Dividing by 3^n :

$$An - A(n-1) = 2 \Rightarrow A = 2.$$

Step 4 — general solution. $a_n = a_n^h + a_n^p = A_1 3^n + 2n 3^n$.

Final answer. $a_n = (A_1 + 2n) 3^n$ (where A_1 is fixed by any given initial condition).

Q24. (7 marks) Use the generating function to solve $a_n = 2a_{n-1} + 1$, $a_0 = 0$.

Solution — full explanation.

Step 1 — set up. Let $G(x) = \sum_{n \geq 0} a_n x^n$. Multiply by x^n , sum for $n \geq 1$:

$$\sum_{n \geq 1} a_n x^n = 2 \sum_{n \geq 1} a_{n-1} x^n + \sum_{n \geq 1} x^n.$$

Step 2 — rewrite. $\sum_{n \geq 1} a_n x^n = G - a_0 = G$ (since $a_0 = 0$); $\sum a_{n-1} x^n = xG$; $\sum_{n \geq 1} x^n = \frac{x}{1-x}$.
So

$$G = 2xG + \frac{x}{1-x} \Rightarrow G(1-2x) = \frac{x}{1-x} \Rightarrow G = \frac{x}{(1-x)(1-2x)}.$$

Step 3 — partial fractions. $\frac{x}{(1-x)(1-2x)} = \frac{A}{1-x} + \frac{B}{1-2x}$ with $A(1-2x) + B(1-x) = x$.

- $x = 1$: $A(1-2) = 1 \Rightarrow A = -1$.
- $x = \frac{1}{2}$: $B(1 - \frac{1}{2}) = \frac{1}{2} \Rightarrow B = 1$.

So $G = \frac{-1}{1-x} + \frac{1}{1-2x}$.

Step 4 — read coefficients. $a_n = -1 \cdot 1 + 1 \cdot 2^n = 2^n - 1$.

Final answer. $a_n = 2^n - 1$. (Check: $a_0 = 1 - 1 = 0$ ✓, $a_1 = 2(0) + 1 = 1 = 2 - 1$ ✓.)

Q25. (7 marks) Solve the recurrence $a_n = 3a_{n-1} - 2a_{n-2}$ with $a_0 = 2$, $a_1 = 5$.

Solution — full explanation.

What the question asks: solve a second-order homogeneous recurrence with two initial conditions.

Step 1 — characteristic equation. Rewrite as $a_n - 3a_{n-1} + 2a_{n-2} = 0$ and solve

$$\alpha^2 - 3\alpha + 2 = 0 \Rightarrow (\alpha - 1)(\alpha - 2) = 0 \Rightarrow \alpha = 1, 2.$$

Step 2 — general solution (distinct roots). For distinct roots, $a_n = A\alpha_1^n + B\alpha_2^n$:

$$a_n = A \cdot 1^n + B \cdot 2^n = A + B \cdot 2^n.$$

Step 3 — use the initial conditions.

$$a_0 = A + B = 2, \quad a_1 = A + 2B = 5.$$

Subtract the first from the second: $B = 3$. Then $A = 2 - B = -1$.

Step 4 — answer. $a_n = -1 + 3 \cdot 2^n$.

Final answer. $a_n = 3 \cdot 2^n - 1$. (Check: $a_0 = 3 - 1 = 2$ ✓, $a_1 = 6 - 1 = 5$ ✓.)

Q26. (7 marks) Solve $a_r - 3a_{r-1} + 2a_{r-2} = 0$ ($r \geq 2$), $a_0 = 2$, $a_1 = 3$, using the generating function.

Solution — full explanation.

Step 1 — set up. Let $G(x) = \sum_{r \geq 0} a_r x^r$. Multiply by x^r , sum for $r \geq 2$:

$$\sum_{r \geq 2} a_r x^r - 3 \sum_{r \geq 2} a_{r-1} x^r + 2 \sum_{r \geq 2} a_{r-2} x^r = 0.$$

Step 2 — rewrite each sum.

$$\sum_{r \geq 2} a_r x^r = G - 2 - 3x, \quad \sum_{r \geq 2} a_{r-1} x^r = x(G - 2), \quad \sum_{r \geq 2} a_{r-2} x^r = x^2 G.$$

So

$$(G - 2 - 3x) - 3x(G - 2) + 2x^2 G = 0.$$

Step 3 — solve for G . Expanding: $G - 2 - 3x - 3xG + 6x + 2x^2 G = 0 \Rightarrow G(1 - 3x + 2x^2) = 2 - 3x$. Since $1 - 3x + 2x^2 = (1 - x)(1 - 2x)$:

$$G = \frac{2 - 3x}{(1 - x)(1 - 2x)}.$$

Step 4 — partial fractions. $\frac{2 - 3x}{(1 - x)(1 - 2x)} = \frac{A}{1 - x} + \frac{B}{1 - 2x}$ with $A(1 - 2x) + B(1 - x) = 2 - 3x$.

- $x = 1$: $A(1 - 2) = 2 - 3 = -1 \Rightarrow A = 1$.
- $x = \frac{1}{2}$: $B(1 - \frac{1}{2}) = 2 - \frac{3}{2} = \frac{1}{2} \Rightarrow B = 1$.

So $G = \frac{1}{1 - x} + \frac{1}{1 - 2x}$.

Step 5 — read coefficients. $a_r = 1 + 2^r$.

Final answer. $a_r = 1 + 2^r$. (Check: $a_0 = 1 + 1 = 2 \checkmark$, $a_1 = 1 + 2 = 3 \checkmark$.)

Q27. (7 marks) Solve $a_n = a_{n-1} + a_{n-2} + 3n + 1$ for $n \geq 2$, with $a_0 = 2$, $a_1 = 3$.

Solution — full explanation.

What the question asks: solve a non-homogeneous recurrence whose right side is a linear polynomial $3n + 1$.

Step 1 — homogeneous part. Solve $a_n - a_{n-1} - a_{n-2} = 0$:

$$\alpha^2 - \alpha - 1 = 0 \Rightarrow \alpha = \frac{1 \pm \sqrt{5}}{2}.$$

These are distinct, so $a_n^h = C_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + C_2 \left(\frac{1-\sqrt{5}}{2} \right)^n$.

Step 2 — particular part. Since $f(n) = 3n + 1$ is a polynomial of degree 1, assume $a_n^p = A + Bn$. Then $a_{n-1}^p = A + B(n-1)$, $a_{n-2}^p = A + B(n-2)$. Substitute:

$$(A + Bn) - (A + Bn - B) - (A + Bn - 2B) = 3n + 1.$$

Simplify the left side: $A + Bn - A - Bn + B - A - Bn + 2B = -A - Bn + 3B$. So

$$-Bn + (3B - A) = 3n + 1.$$

Match coefficients: $-B = 3 \Rightarrow B = -3$; and $3B - A = 1 \Rightarrow -9 - A = 1 \Rightarrow A = -10$. So $a_n^p = -10 - 3n$.

Step 3 — general solution and constants.

$$a_n = C_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + C_2 \left(\frac{1-\sqrt{5}}{2} \right)^n - 3n - 10.$$

Using $a_0 = 2$: $C_1 + C_2 - 10 = 2 \Rightarrow C_1 + C_2 = 12$. Using $a_1 = 3$: $C_1 \alpha_1 + C_2 \alpha_2 - 13 = 3 \Rightarrow C_1 \alpha_1 + C_2 \alpha_2 = 16$. Solving gives $C_1 = 6 + 2\sqrt{5}$, $C_2 = 6 - 2\sqrt{5}$.

Final answer. $a_n = (6 + 2\sqrt{5}) \left(\frac{1+\sqrt{5}}{2} \right)^n + (6 - 2\sqrt{5}) \left(\frac{1-\sqrt{5}}{2} \right)^n - 3n - 10$.

Q28. (7 marks) Solve $a_n - 2a_{n-1} - 3a_{n-2} = 0$ with $a_0 = 3$, $a_1 = 1$, using the generating function.

Solution — full explanation.

Step 1 — set up. Let $G = \sum_{n \geq 0} a_n x^n$; multiply by x^n and sum for $n \geq 2$:

$$(G - 3 - x) - 2x(G - 3) - 3x^2G = 0.$$

Step 2 — solve for G . Expanding: $G - 3 - x - 2xG + 6x - 3x^2G = 0 \Rightarrow G(1 - 2x - 3x^2) = 3 - 5x$. Since $1 - 2x - 3x^2 = (1 - 3x)(1 + x)$:

$$G = \frac{3 - 5x}{(1 - 3x)(1 + x)}.$$

Step 3 — partial fractions. $\frac{3 - 5x}{(1 - 3x)(1 + x)} = \frac{A}{1 - 3x} + \frac{B}{1 + x}$ with $A(1 + x) + B(1 - 3x) = 3 - 5x$.

- $x = -1$: $B(1 + 3) = 3 + 5 = 8 \Rightarrow B = 2$.
- $x = \frac{1}{3}$: $A(1 + \frac{1}{3}) = 3 - \frac{5}{3} = \frac{4}{3} \Rightarrow \frac{4}{3}A = \frac{4}{3} \Rightarrow A = 1$.

$$\text{So } G = \frac{1}{1 - 3x} + \frac{2}{1 + x}.$$

Step 4 — read coefficients. Using $\frac{1}{1 + x} = \sum (-1)^n x^n$:

$$a_n = 3^n + 2(-1)^n.$$

Final answer. $a_n = 3^n + 2(-1)^n$. (Check: $a_0 = 1 + 2 = 3 \checkmark$, $a_1 = 3 - 2 = 1 \checkmark$.)

Q29. (7 marks) Solve $a_n - 6a_{n-1} + 9a_{n-2} = 0$ for all $n \geq 2$, using the generating function.

Solution — full explanation.

Step 1 — set up. Let $G = \sum_{n \geq 0} a_n x^n$; multiply by x^n and sum for $n \geq 2$:

$$(G - a_0 - a_1x) - 6x(G - a_0) + 9x^2G = 0.$$

Step 2 — solve for G . Collecting G : $G(1 - 6x + 9x^2) = a_0 + (a_1 - 6a_0)x$. Since $1 - 6x + 9x^2 = (1 - 3x)^2$:

$$G = \frac{a_0 + (a_1 - 6a_0)x}{(1 - 3x)^2}.$$

Step 3 — read coefficients. Use the standard series

$$\frac{1}{(1 - 3x)^2} = \sum_{n \geq 0} (n + 1) 3^n x^n, \quad \frac{x}{(1 - 3x)^2} = \sum_{n \geq 1} n 3^{n-1} x^n.$$

Hence, for $n \geq 1$,

$$a_n = a_0(n + 1)3^n + (a_1 - 6a_0)n 3^{n-1}.$$

Step 4 — general form. Because the root $\alpha = 3$ is repeated, the general solution is

$$a_n = (A + Bn)3^n,$$

with $A = a_0$ and B fixed by a_1 . (No initial conditions were given, so we leave it in this general form.)

Final answer. $a_n = (A + Bn)3^n$.

Q30. (7 marks) Find the sequence b_r from the generating function $\sum b_r x^r = \frac{1}{x^2 - 5x + 6}$.

Solution — full explanation.

What the question asks: given $G(x)$, find the coefficient b_r of x^r .

Step 1 — factorise the denominator. $x^2 - 5x + 6 = (x - 2)(x - 3)$, so $G(x) = \frac{1}{(x - 2)(x - 3)}$.

Step 2 — partial fractions. $\frac{1}{(x - 2)(x - 3)} = \frac{A}{x - 2} + \frac{B}{x - 3}$ with $A(x - 3) + B(x - 2) = 1$.

- $x = 2$: $A(2 - 3) = 1 \Rightarrow A = -1$.
- $x = 3$: $B(3 - 2) = 1 \Rightarrow B = 1$.

$$\text{So } G(x) = \frac{-1}{x - 2} + \frac{1}{x - 3} = \frac{1}{2 - x} - \frac{1}{3 - x}.$$

Step 3 — expand each term.

$$\frac{1}{2 - x} = \frac{1}{2} \cdot \frac{1}{1 - x/2} = \sum_{r \geq 0} \frac{x^r}{2^{r+1}}, \quad \frac{1}{3 - x} = \sum_{r \geq 0} \frac{x^r}{3^{r+1}}.$$

Step 4 — read the coefficient of x^r .

$$b_r = \frac{1}{2^{r+1}} - \frac{1}{3^{r+1}} = \frac{3^{r+1} - 2^{r+1}}{6^{r+1}}.$$

Final answer. $\boxed{b_r = \frac{1}{2^{r+1}} - \frac{1}{3^{r+1}}}$. (Check: $b_0 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} = G(0)$ ✓.)

Q31. (7 marks) Solve $a_{n+2} - 4a_{n+1} + 4a_n = 2^n$.

Solution — full explanation.

What the question asks: solve a non-homogeneous recurrence where the right side equals a repeated characteristic root's power.

Step 1 — homogeneous part. Solve $a_{n+2} - 4a_{n+1} + 4a_n = 0$:

$$\alpha^2 - 4\alpha + 4 = (\alpha - 2)^2 = 0 \Rightarrow \alpha = 2 \text{ (double root)}.$$

So $a_n^h = (A + Bn)2^n$.

Step 2 — choose the particular form. The right side is 2^n , so $\beta = 2$, which is a **double** characteristic root. Each time β is a root we multiply by an extra factor of n ; since the multiplicity is 2, we multiply by n^2 :

$$a_n^p = C n^2 2^n.$$

Step 3 — substitute and solve for C . Put $a_n^p = C n^2 2^n$, $a_{n+1}^p = C(n + 1)^2 2^{n+1}$, $a_{n+2}^p = C(n + 2)^2 2^{n+2}$:

$$C(n + 2)^2 2^{n+2} - 4C(n + 1)^2 2^{n+1} + 4C n^2 2^n = 2^n.$$

Divide by 2^n : $4C(n + 2)^2 - 8C(n + 1)^2 + 4C n^2 = 1$. Factor $4C$:

$$4C[(n + 2)^2 - 2(n + 1)^2 + n^2] = 1.$$

The bracket equals 2 (all the n terms cancel). So $4C \cdot 2 = 1 \Rightarrow 8C = 1 \Rightarrow C = \frac{1}{8}$.

Step 4 — general solution.

$$a_n = (A + Bn)2^n + \frac{1}{8}n^2 2^n = \left(A + Bn + \frac{n^2}{8}\right)2^n.$$

Final answer. $\boxed{a_n = \left(A + Bn + \frac{n^2}{8}\right)2^n}$.

Q32. (7 marks) In how many ways can 5 people be seated in a row if two of them, Alice and Bob, must sit next to each other?

Solution — full explanation.

What the question asks: count line seatings of 5 with Alice and Bob adjacent.

The block idea. Tie Alice and Bob into one block. Then we are arranging **4 units**: the block plus the other 3 people.

Arrange the units. The 4 units can be ordered in $4! = 24$ ways.

Arrange inside the block. Alice and Bob can sit in either order (AB or BA): $2! = 2$ ways.

Combine (product rule).

$$4! \times 2! = 24 \times 2 = 48.$$

Final answer. 48 ways.

Quick pointers for Module 2: the most repeated items are *solving recurrence relations* (by the characteristic-equation method or generating functions), *counting with inclusion–exclusion*, and *arrangement problems* (together / never-adjacent / circular). Generating-function solutions of recurrences appear as full 7-mark questions very often.